

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science **ASSESSMENT**

TOOL 3 – Class Test

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 3 – Class Test
Weightage:	20% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	A. Mounish	Reg. No.:	192472273
Section / Batch:	CSE-AI	Date:	29/07/26

Code of Conduct

*I **A. Mounish / 192472273** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
N-gram Models and Word Counting for Language Modeling	N-gram Construction, Word Frequency Analysis, Probability Estimation	1
Unsmoothed N-grams and Smoothing Techniques in Syntax Analysis	Unsmoothed N-grams, Backoff, Deleted Interpolation	2
Entropy and Language Uncertainty in NLP Systems	Information Theory, Entropy Measurement, Language Prediction	3
English Word Classes and Tagsets for Linguistic Analysis	Word Categories, POS Categories, English Tagsets	4
Part-of-Speech Tagging Techniques and Applications	POS Tagging, Rule-Based Tagging, Stochastic Tagging, Transformation-Based Tagging	5

Annexure A – Class Test

1. A document processing system encounters the words “talked”, “talking”, and “talks”. Apply inflectional morphology rules and Porter Stemmer steps to derive the normalized root form. Identify the suffixes removed and explain the role of morphological normalization in improving information retrieval accuracy.
2. Given the input words: “relational”, “conditional”, “happiness”, “running”, design a hybrid morphological processing pipeline using derivational rules, inflectional analysis, finite-state morphological parsing, and Porter Stemmer. Decompose each word into root and affixes, represent transitions using finite-state parsing logic, and apply stemming steps to obtain the final normalized forms. Analyze how each technique contributes to accurate root extraction.
3. An enterprise-level search engine optimization (SEO) system processes web content containing variations such as “optimize”, “optimization”, “optimized”, and “optimizing”. The system must ensure consistent keyword indexing and ranking. Develop a morphology-driven solution integrating inflectional and derivational analysis, finite-state morphological parsing for rule validation, and Porter Stemmer for normalization. Apply the approach to the given inputs and determine how the system achieves improved search relevance and reduced redundancy.
4. A large-scale information retrieval system processes input words: “compute”, “computer”, “computing”, and “computation” to enhance search accuracy.
Design a hybrid morphological pipeline integrating inflectional and derivational analysis, finite-state morphological parsing, and Porter Stemmer.
Decompose each word into root and affixes, represent transformations using state transitions, and apply stemming rules for normalization.
Ensure consistent root extraction across all variations for efficient indexing.
Determine the morphological breakdown, parsing transitions, and final normalized forms for each word.
5. A text mining system processes input words: “decide”, “decision”, “decisive”, and “deciding” to support semantic clustering and analysis.
Construct a morphology-based framework combining derivational rules, inflectional transformations, finite-state parsing, and Porter stemming.
Analyze how suffix variations affect meaning and grammatical structure while maintaining root consistency. Apply the approach to extract roots, validate transformations, and normalize the dataset.
Derive the morphological structure, state transitions, and final stem for each input word.

Q1. Inflectional Morphology and Porter Stemmer

Input Words

talked, talking, talks

Step 1: Morphological Decomposition

Word Root Suffix

talked talk -ed

talking talk -ing

talks talk -s

Step 2: Inflectional Analysis

Word Type

talked Past tense (Inflectional)

talking Present participle (Inflectional)

talks Third-person singular (Inflectional)

Step 3: Porter Stemmer

- talked → remove -ed → talk
- talking → remove -ing → talk
- talks → remove -s → talk

Final Normalized Form

Word Normalized Root

talked talk

talking talk

talks talk

Role of Morphological Normalization

All grammatical forms are mapped to talk, improving search accuracy, reducing duplicate entries, and increasing information retrieval efficiency.

Q2. Hybrid Morphological Processing Pipeline

Input Words

relational, conditional, happiness, running

Step 1: Morphological Decomposition

Word Root Affix

relational relate -ional

conditional condition -al

happiness happy -ness

running run -ing

Step 2: Inflectional / Derivational Analysis

Word	Type
relational	Derivational
conditional	Derivational
happiness	Derivational
running	Inflectional

Step 3: Finite-State Parsing

relate → relational
condition → conditional
happy → happiness
run → running

Step 4: Porter Stemmer

Word	Stem
relational	relat
conditional	condit
happiness	happi
running	run

Final Normalized Forms

Word	Final Stem
relational	relat
conditional	condit
happiness	happi
running	run

Contribution of Each Technique

- **Derivational analysis:** Identifies word formation.
- **Inflectional analysis:** Handles grammatical changes.
- **Finite-state parsing:** Validates transformations.
- **Porter Stemmer:** Produces a common normalized stem.

Q3. Morphology-Driven SEO Solution

Input Words

optimize, optimization, optimized, optimizing

Step 1: Morphological Decomposition

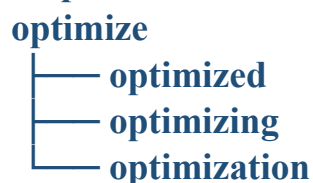
Word	Root	Affix
optimize	optimize	—
optimization	optimize	-ation
optimized	optimize	-ed

Word	Root	Affix
optimizing	optimize	-ing

Step 2: Classification

Word	Type
optimize	Base
optimization	Derivational
optimized	Inflectional
optimizing	Inflectional

Step 3: Finite-State Parsing



Step 4: Porter Stemmer

Word	Final Stem
optimize	optim
optimization	optim
optimized	optim
optimizing	optim

Final Normalized Form

All words → optim

Result

The search engine indexes all forms under one keyword, improving search relevance and reducing redundant indexing.

Q4. Hybrid Morphological Pipeline

Input Words

compute, computer, computing, computation

Step 1: Morphological Decomposition

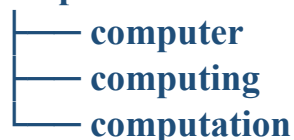
Word	Root	Affix
compute	compute	—
computer	compute	-er
computing	compute	-ing
computation	compute	-ation

Step 2: Classification

Word	Type
compute	Base
computer	Derivational
computing	Inflectional
computation	Derivational

Step 3: Finite-State Parsing

compute



Step 4: Porter Stemmer

Word	Final Stem
compute	comput
computer	comput
computing	comput
computation	comput

Final Normalized Form

All words → comput

Q5. Morphology-Based Framework

Input Words

decide, decision, decisive, deciding

Step 1: Morphological Decomposition

Word	Root	Affix
decide	decide	—
decision	decide	-sion
decisive	decide	-ive
deciding	decide	-ing

Step 2: Classification

Word	Type
decide	Base
decision	Derivational
decisive	Derivational
deciding	Inflectional

Step 3: Finite-State Parsing

decide

— decision
 — decisive
 — deciding

Step 4: Porter Stemmer

Word Final Stem

decide decis

decision decis

decisive decis

deciding decis

Final Normalized Form

All words → decis

Effect of Suffix Variations

- **-sion forms a noun.**
- **-ive forms an adjective.**
- **-ing indicates present participle.**
- **All forms are normalized to the common stem decis for semantic clustering and efficient retrieval.**

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions

Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time
--------------------------------	-----	---	---	----------------------------------	----------------------------

Q. No. / Criteria Level	Understanding of Task	Procedure & Execution	Observation & Result	Viva Voce / Conceptual Response	Time Management & Presentation	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____